Counting:

Multiplication rule:

Sets A, B

$$A \times B = \{(a,b) : a \in A, b \in B\}$$

$$|A \times B| = |A|\,|B|$$

$$A_1 \subseteq A, \quad B_1 \subseteq B$$

$$A_1 \times B_1 = \{(a,b) : a \in A_1, b \in B_1\} \subseteq A \times B$$

$$|A_1 \times B_1| = |A_1| \cdot |B_1|$$

Ex: Soup (5), Starters (20), Orders (soup, starter): $5 \times 20 = 100$

Addition rule: A, B: disjoint

$$|A \cup B| = |A| + |B|$$

Ordered subsets

$[n] \triangleq \{1, 2, \ldots, n\}$

$[n]$: size k subsets

Ex: $[3]$

Subsets of size 2: $\{1,2\}, \{2,3\}, \{1,3\}$

$$\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$$ Unordered

Ordered subsets of size 2: $\{1,2\}$ $\{2,3\}$ $\{1,3\}$

$\{2,1\}$ $\{3,2\}$ $\{3,1\}$

$$n(n-1)\cdots(n-(k-1)) = \frac{n!}{(n-k)!}$$ ordered

Ex: ① 120 boys 30 girls

Committee 6 boys, 4 girls

of ways $\binom{120}{6} \cdot \binom{30}{4}$

② Donut shop

10 varieties (inexhaustible supply)

12 in a box

ordered: 10^{12}

unordered: $\binom{12+10-1}{10-1} = \binom{12+10-1}{12}$

new → # non-negative integer sols of

$a_1 + a_2 + \cdots + a_{10} = 12$

③ $(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$

$(x+y)(x+y)\cdots(x+y)$

$\sum_{i=0}^{n} \binom{n}{i} = 2^n$

$\binom{n}{i} = \binom{n}{n-i}$

Multinomial: $[n]$

→ Pick 1 subset of size k $\binom{n}{k}$ possibilities

Pick 1 subset of size k_1 & a second subset of size k_2

$$\binom{n}{k_1} \cdot \binom{n-k_1}{k_2} = \frac{n!}{k_1!\, k_2!\, (n-k_1-k_2)!}$$

General: $n = n_1 + n_2 + \cdots + n_k$

Split $[n]$ into ℓ subsets of size $n_1, \dots, n_\ell$

$$\#\text{ ways} = \binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots$$

$$= \frac{n!}{n_1! \cdot n_2! \cdots n_\ell!} \triangleq \binom{n}{n_1, n_2, \dots, n_\ell}$$

$$(x_1 + x_2 + \cdots + x_\ell)^n = \sum_{\substack{n_1, \dots, n_\ell : \\ n_1 + \cdots + n_\ell = n}} \binom{n}{n_1, \dots, n_\ell} x_1^{n_1} x_2^{n_2} \cdots x_\ell^{n_\ell}$$

Ex: Card games

(1) 52 cards dealt to 4 players (13 cards each)

$$\binom{52}{13, 13, 13, 13} = \frac{52!}{(13!)^4}$$

Counting unions → Inclusion-exclusion principle.

10 balls, 4 bins

